16. Why are NOR and NAND gates called universal gates?
17. Define barrier potential.
18. What is rectification?
19. List the applications of light emitting diode.
20. Give the principle of solar cells.
21. What is an integrated circuit?
22. What is modulation?
23. Define bandwidth of transmission system.
24. What do you mean by skip distance?
25. Give applications of RADAR.
26. What is mobile communication?
27. Explain centre frequency or resting frequency in frequency modulation.
28. What does RADAR stand for?
29. Fiber optic communication is gaining popularity among the various transmission media -justify.

## III. Long Answer Questions

1. Elucidate the formation of n-type extrinsic semiconductors.
2. Explain the formation of depletion region and barrier potential in PN junction diode.
3. Draw the circuit diagram of a half wave rectifier and explain its working.
4. Explain the construction and working of a full wave rectifier.
5. What is an LED? Give the principle of its operation with a diagram.
6. Write a note on photodiode.
7. Explain the working principle of a solar cell. Mention its applications.
8. Sketch the static characteristics of a common emitter transistor and bring out the essential features of input and output characteristics.
9. Transistor functions as a switch. Explain.
10. Describe the function of a transistor as an amplifier with the neat circuit diagram. Sketch the input and output wave forms.
11. Give circuit symbol, logical operation, truth table, and Boolean expression of i) AND gate ii) OR gate iii) NOT gate iv) NAND gate v) NOR gate and vi) EX-OR gate.
12. State and prove De Morgan's first and second theorem.
13. Explain the ampitude modulation with necessary diagrams.
14. Explain the basic elements of communication system with the necessary block diagram.
15. Explain the ground wave propagation and space wave propagation of electromagnetic waves through space.
16. List out the advantages and limitations of frequency modulation.
17. What is meant by satellite communication? Give its applications.

## IV. Numerical Problems

1. The given circuit has two ideal diodes connected as shown in figure below. Calculate the current flowing through the resistance $R_1$. [Ans: 2.5 A]



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